Ning Yang

Ph.D. Candidate, Expected Dec 2026

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State Key Laboratory of Geomechanics and Geotechnical Engineering Safety, Institute of Rock and Soil Mechanics, Chinese Academy of Sciences, Wuhan, Hubei, China

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Research Interests

I am deeply passionate about exploring the natural world through a multidisciplinary lens, from physics, earth sciences, and engineering. My recent work centers on the following key research questions:

- **Earthquake Physics:** How do earthquakes arise from fault slip?
- **Fault Friction:** How can rate-and-state friction be extended to micro-scale grain mechanisms?
- **Fluid-Induced Seismicity:** How to tell induced earthquakes apart from natural earthquakes?
- **Model Translation**: How to apply fault-physics models for practical engineering use?
- **AI-Augmented Geoscience**: How can AI advance fault and induced-seismicity research?

Education

2023.09–2026.12 **Ph.D. Civil Engineering**, Institute of Rock and Soil Mechanics, Chinese Academy of Sciences (IRSM, CAS), Wuhan, China
Advisor: Peng-Zhi Pan, Professor
Committee: To be determined
Thesis: Mechanisms of fault frictional healing on instability and seismicity under fluid injection

2020.09–2023.06 **M.S. Hydraulic Engineering**, Qingdao University of Technology, Qingdao, China
Advisor: Xiaoming Guan, Associate Professor
Committee: Yongyan Wang, Chongge Wang, Xuefeng Wu, Liang Kong, Fanxiu Chen, Liang Li, Sulei Zhang
Thesis: Study on damage characteristics and vibration control of large diameter pipelines affected by blasting construction of close distance tunnel

2016.09–2020.07 **B.S. Civil Engineering**, Qingdao University of Technology, Qingdao, China

Peer-Reviewed Publications

[8] **Ning Yang**, Peng-Zhi Pan, et al. Mechanism of fault healing and slip nucleation under thermo-hydro-mechanical coupling. International *Journal of Rock Mechanics and Mining Sciences*. (In preparation)

[7] **Ning Yang**, Peng-Zhi Pan, et al. Frictional healing-regulated stability of injection-induced fault reactivation. *Earth and Planetary Science Letters*. (Submitted)

[6] Peng-Zhi Pan, **Ning Yang**, et al. Fault healing and its control on reactivation: from laboratory experiments to subsurface energy applications. *Renewable and Sustainable Energy Reviews*. (Under review)

[5] **Ning Yang**, Peng-Zhi Pan*, et al. Laboratory-scale characteristics of micro-slip during fault quiescence and implications for injection-induced reactivation stability, *International Journal of Rock Mechanics and Mining Sciences*, 2026, 197, 106339. (IF 8.5, JCR Q1, Link)

[4] Xiaoming Guan, **Ning Yang***, et al. Vibration response and failure modes analysis of the

temporary support structure under blasting excavation of tunnels, *Engineering Failure Analysis*, 2022, 136, 606188. (IF 7.0, JCR Q1, Link)

[3] Xiaoming Guan, **Ning Yang**, et al. Dynamic Response and Improved Safety Prediction Method for Large Diameter Pipelines with Different Shapes Subjected to Close-Proximity Tunnel Blasting Vibration, *Soil Dynamics and Earthquake Engineering*, 2026, 209, 110448. (IF 5.1, JCR Q1, Link)

[2] **Ning Yang**, Xin Jin, et al. Driving Parameter Trends and a Safety Evaluation Method Based on the Composite Ratio for Shield Tunneling in Complex Strata. ASCE-ASME *Journal of Risk and Uncertainty in Engineering Systems, Part A: Civil Engineering*, 2023,9(2), 04023007. (IF 2.9, JCR Q2, Link)

[1] Xiaoming Guan, Yingkang Yao, **Ning Yang**, et al. Analysis of factors influencing vibration reduction and design optimization of damping holes in adjacent tunnel blasting. *Case Studies in Construction Materials*, 2023, 19, e02448. (IF 6.9, JCR Q1, Link)

Issued Chinese Invention Patents

[6] Xiaoming Guan, **Ning Yang**, et al. Blasting vibration reduction method for nearby tunnels with auxiliary pilot tunnels and pre-set composite vibration isolation layers. CN112539064B. (Link)

[5] Xiaoming Guan, **Ning Yang**, et al. Multi-parameter coupled quantitative evaluation of interlayer grouting reinforcement in stacked tunnels. CN112417367B. (Link)

[4] Jianyong An, **Ning Yan**g, et al. Deep open dewatering technique for tunnels in water-rich composite strata. CN113217097B. (Link)

[3] Xiaoming Guan, **Ning Yang**, et al. Blasting vibration safety assessment for buildings. CN114781040B. (Link)

[2] Xiaoming Guan, **Ning Yang**, et al. Blasting charge calculation for near-field protected structures. CN114936342B. (Link)

[1] Xiaoming Guan, **Ning Yang**, et al. Optimization method for deep-hole blasting. CN115600368B. (Link)

First-Authored Conference Presentations

[3] **Ning Yang**, Peng-Zhi Pan. Thermal healing regulates injection-induced fault reactivation stability. *CouFrac2026*, September 20–23, Uppsala, Sweden. (Oral)

[2] **Ning Yang**, Peng-Zhi Pan, et al. Preliminary investigation on micro-slip and pore-pressure perturbation reactivation mechanism of faults during healing periods in waste repositories. *The 9th Symposium on Underground Waste Disposal, Chinese Society for Rock Mechanics and Engineering*, Aug 23, 2025, Guiyang, China. (Oral; Recommended Young Investigator Presentation)

[1] **Ning Yang**, Peng-Zhi Pan. Numerical study on friction weakening and re-nucleation of faults induced by fluid injection during the interseismic period. *CouFrac2024*, Nov 13–15, Kyoto, Japan. (Poster; Invited to submit a full journal paper)

Honors and Awards

2026.06 **Merit Student**, IRSM, CAS

2023.03 **Outstanding Graduate of Shandong Province**, Shandong Provincial Department of Education (Link)

2022.11 **National Scholarship**, Ministry of Education of P.R. China (Link)

Presented to top-performing postgraduate students based on academic excellence.

2023.04 **Second Prize, Science and Technology Award**, China Communications and Transportation Association (Link)
Project: Key technology research and application of blasting vibration control for tunnels in complex urban environments, Rank 7/15

2020.06 **Excellent Graduation Thesis**, Qingdao University of Technology

2020.06 **Selected for Public Defense of Undergraduate Graduation Thesis**, Qingdao University of Technology, Rank 4/440 (Link)
Competitive oral defense reserved for top undergraduate theses, reviewed by a panel of academic and industry experts from the School of Civil Engineering. Only a small fraction of the 440 graduating students were invited to this public presentation.

Academic Exchange and Training

2023.09–2026.12 Graduate Research Assistant State Key Laboratory of Geomechanics and Geotechnical Engineering Safety, IRSM, CAS Advisor: **Peng-Zhi Pan**
• Investigated cyclic evolution behaviors of faults under fluid injection perturbation.
• Performed shear-flow coupled laboratory tests and rate-and-state friction based numerical simulations to reveal the thermo-hydro-mechanical (THM) coupling mechanism of fault healing, stability evolution and earthquake event scaling.
• Transferred findings to applications in geological energy development and induced seismic hazard assessment.

2026.08–2026.08 Academic Visit Coordination Mentor: **Junlong Shang** (University of Manchester, UK) Invited by the State Key Laboratory of Geomechanics and Geotechnical Engineering, hosted by Prof. Peng-Zhi Pan
• Provided full-cycle logistical support and assisted academic discussions.
• Seminar covered fracture heterogeneity of subsurface rock systems from geometric, mineralogical and hydro-mechanical perspectives.
• Discussed implications for geothermal exploitation, energy storage, nuclear waste disposal and induced seismicity risk management for low-carbon subsurface projects.

2025.09–2025.09 Academic Visit Coordination Mentor: **Arno Zang** (GFZ Helmholtz Centre for Geosciences, Germany) Invited by the State Key Laboratory of Geomechanics and Geotechnical Engineering, hosted by Prof. Peng-Zhi Pan
• Provided full-cycle logistical support and assisted academic discussions.
• Seminar introduced rock hydraulic-fatigue mechanisms and fatigue hydraulic fracturing under cyclic pulsed fluid injection.
• Explored adaptive injection strategies to mitigate induced seismicity.

2025.09–2025.09 Academic Visit Coordination Mentor: **Qingyun Duan** (Hohai University, China) Invited by Prof. Peng-Zhi Pan, Ning Yang
• Provided full-cycle logistical support and assisted academic discussions.
• Seminar shared practical guidelines for preparing high-quality review manuscripts for top geoscience journals, with focus on the invitation-only Reviews of Geophysics (AGU).
• Covered review paper classification, manuscript structure, proposal writing, submission workflow and common pitfalls in peer review.

2025.05–2025.06 Special Training on AI-enabled Underground Engineering, **Chinese Academy of Sciences**, Beijing

• Investigated data-mechanism hybrid models including Physics-Informed Neural Networks (PINN) and Graph Neural Networks (GNN), and analyzed their challenges in geohazard prediction to balance physical interpretability and prediction accuracy.

2024.11–2024.11 Academic Visit Coordination Mentor: **Amin Hekmatnejad** (Pontificia Universidad Católica de Valparaíso, Chile) Invited by the State Key Laboratory of Geomechanics and Geotechnical Engineering, hosted by Prof. Peng-Zhi Pan

• Provided full-cycle logistical support and assisted academic discussions.

• Seminar presented Universal Discontinuity Index (UDI) and photogrammetry-deep learning workflows for automated rock mass assessment in underground engineering.

2025.11–2025.11 Academic Visit Coordination Mentor: **Christopher Spiers** (Utrecht University, Netherlands) Invited by the State Key Laboratory of Geomechanics and Geotechnical Engineering, hosted by Prof. Peng-Zhi Pan

• Provided full-cycle logistical support and assisted academic discussions.

• Seminar covered micro-physical mechanisms of rock salt deformation-damage-healing for geological energy/waste storage.

• Discussed physical nature of fault friction and modeling strategies for natural- and injection-induced earthquakes under crustal thermo-hydro-mechanical conditions.

Research and Field Experience

2026.03–2026.12 **Open Fund of State Key Laboratory** (SKLGGES-024007): Injection-induced seismicity and associated source mechanisms: Insights from lab tests and micro-DEM modelling

• Key project participant

• Coordinated access to the multi-field shear-rheology rock testing apparatus and supported lab-scale experimental work, including hands-on experimental training.

2024.07–2026.12 **National Science Fund for Distinguished Young Scholars** (52125903): Failure mechanism and analytical methods for engineering rock masses

• Core team member

• Performed laboratory experiments and theoretical studies focusing on shear-friction stability of rock faults.

2023.07–2026.05 Study on surrounding-rock stability, seepage evaluation and support optimization for underground caverns of Luxi Pumped-Storage Hydropower Station, **PowerChina Kunming Engineering Corp., Ltd.**, Yunnan, China (E3415201)

• Project lead

• Led numerical analysis, report preparation, project defense, acceptance review and budget administration.

• Evaluated surrounding-rock and seepage stability, and proposed optimized support strategies for large-scale underground cavern complexes.

2024.11–2024.11 Acceptance test and technical training for in-situ Computed Tomography (CT) experimental facility, **Sanying Precision Instruments Co., Ltd**., Tianjin, China

• Completed technical training covering working principles and operational workflows of the in-situ CT rock-testing system.

• Conducted full-system acceptance verification for the experimental apparatus.

2024.03–2024.09 Failure modes, stability criteria and countermeasures for high-wall caverns under complex geological conditions at GS Hydropower Station, **PowerChina Kunming Engineering Corp., Ltd.**, Yunnan, China (E2415201)

• *Lead of dynamic-simulation subgroup*
• *Carried out numerical modelling, technical writing and supported project defense, and final acceptance.*
• *Investigated instability mechanisms and safety performance of layered-jointed rock caverns subjected to seismic disturbance.*

2023.11–2023.11 Field coal-mine investigation with Assoc. Prof. **Peter Konicek** and **Vlastimil Kajzar**, Czech Academy of Sciences, Czech Republic
• *Coordinated itineraries, external communication and on-site logistics.*
• *Participated in underground field surveys at Yima Coal Mine (Mianchi, Henan, China) and Ordos coal-mining sites (Inner Mongolia, China), together with project-oriented academic discussions.*

2022.01–2026.05 Key construction technologies for Zhengzhou (Henan, China) metro crossing major hazard sources, Industrial Project, **China Construction Second Engineering Bureau Ltd.**, Henan, China (B2-2019-0243)
• *Took part in technical research on cut-and-cover separated metro stations within high-water-level, deeply-buried loess strata.*
• *Investigated key construction techniques and risk-mitigation strategies for shield tunnelling in water-rich sandy formations.*

2021.09–2021.11 Big-data-driven green intelligent blasting-control technology for complex subgrades, Industrial Project, **Shandong Hi-Speed Group Co., Ltd.**, Shandong, China (B2-2022-0063)
• *Performed site reconnaissance and blasting-vibration monitoring for cutting-blasting works on the Weifang-Qingdao section of Shandong Expressway.*
• *Contributed to blast-hole pattern design, as well as blasting seismic-wave acquisition and data analysis.*

Fellowships and Grants

2025–2027 **Open Fund of State Key Laboratory** (SKLGGES-024007): Injection-induced seismicity and associated source mechanisms: Insights from lab tests and micro-DEM modelling
PIs: Junlong Shang, Peng-Zhi Pan, **Ning Yang**
•*Contributed to laboratory shear-seepage tests, project coordination and academic visitor arrangements.*

2024–2027 **DECOVALEX 2027, SAFENET-2** (QDS-10139A-CAS), International Research Project: Safety implications of fluid flow, shear, thermal and reaction processes within crystalline rock fracture networks (Link)
PIs: Peng-Zhi Pan, **Ning Yang**, Wenbo Hou
• *Contributed to numerical simulations of fault reactivation and induced seismicity.*

2023–2026 **National Science Fund for Distinguished Young Scholars** (52125903): Fracture mechanism and analysis method of engineering rock masses
PI: Peng-Zhi Pan
• *Contributed to experimental tests, numerical modelling and result interpretation.*

2020–2023 **National Natural Science Foundation of China** (51708317): Study on high-order modal vibration effect and damage characteristics of existing structures under tunnel blasting
PI: Xiaoming Guan
• *Participated in field blasting monitoring and structural damage analysis.*